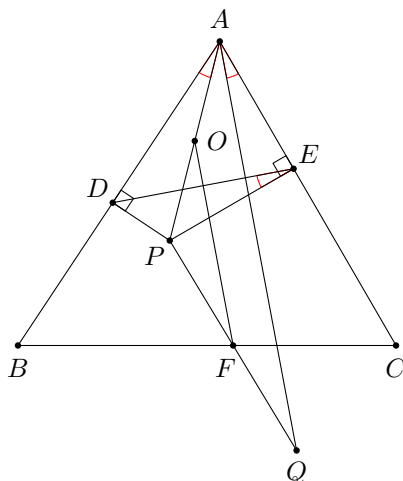


Problem 37 (Baltic Way 2021 Problem 11). Let P be a point inside $\triangle ABC$. Let D and E be the projections of P onto AB and AC respectively. Let F be the point on the line BC such that $DF = EF$. Let Q be the point such that F is the midpoint of PQ . Prove that $\angle BAP = \angle QAC$.



Solution. Note that A, D, P, E are concyclic, and AP is a diameter of $(ADPE)$. Let O be the centre of $(ADPE)$, which means O is the midpoint of AP . By the midpoint theorem, we have $AQ \parallel OF$. As OF is the perpendicular bisector of DE , we obtain $OF \perp DE$, and $AQ \perp DE$. It follows that

$$\angle BAP = \angle DAP = \angle DEP = 90^\circ - \angle AED = \angle QAE = \angle QAC.$$

□

Problem 38 (India MO 2013 Problem 6). Let A and B be two sets of positive real numbers such that $|A| = |B| = 3$. It is known that A and B have the same sum of elements and the same product of elements. Also, $\min A \leq \min B$ and $\max A \geq \max B$ (where $\min A$ means the smallest element in A , etc.). Prove that $A = B$.

Solution. Let $A = \{a, b, c\}$ and $B = \{d, e, f\}$ where $a < b < c$ and $d < e < f$. Define $f(x) = (x - a)(x - b)(x - c)$ and $g(x) = (x - d)(x - e)(x - f)$. Since f and g have the same sum of roots and the same product of roots, we must have $f(x) = g(x) + kx$ for some constant k .

Now, since $0 < a \leq d < e < f$, we have $g(a) = (a - d)(a - e)(a - f) \leq 0 = f(a) = g(a) + ka$, and so $k \geq 0$. Similarly, since $0 < d < e < f \leq c$, we have $g(c) = (c - d)(c - e)(c - f) \geq 0 = f(c) = g(c) + kc$, and so $k \leq 0$. It follows that $k = 0$. This means $f = g$, and so $A = \{a, b, c\} = \{d, e, f\} = B$. □

Problem 39 (British MO 2011 Round 2 Problem 4). Let S be the set of all lattice points (x, y) in the coordinate plane with $1 \leq x, y \leq 2021$. At most how many points can be chosen from S so that there do not exist 4 chosen points forming a non-degenerate parallelogram (in any orientation)?

Solution. The answer is 4041.

We first show that at most 4041 points can be chosen. For each horizontal line $y = c$ where $c = 1, 2, \dots, 2021$, let $f(c)$ be the number of distinct distances between two chosen points on this line. If $k \geq 1$ points are chosen on this line, say $(x_1, c), (x_2, c), \dots, (x_k, c)$ where $x_1 < x_2 < \dots < x_k$, then we have $f(c) \geq k - 1$ since the distances $x_2 - x_1, \dots, x_k - x_1$ are pairwise distinct. Note that $f(c) \geq k - 1$ also holds if $k = 0$.

Now, if the same distance d occurs in two different lines $y = c$ and $y = c'$, then the 4 corresponding points form a non-degenerate parallelogram. Thus, each distance occurs in at most one horizontal line. Also, the distance can only range from 1 to 2020. Therefore, we have

$$2020 \geq \sum_{c=1}^{2021} f(c) \geq \sum_{c=1}^{2021} (k_c - 1) = \sum_{c=1}^{2021} k_c - 2021$$

where k_c is the number of chosen points on the line $y = c$. This implies $k_1 + k_2 + \cdots + k_{2021} \leq 4041$, which means at most 4041 points are chosen.

Lastly, we provide an example of choosing 4041 points. For example, we can choose all points on the lines $x = 1$ and $y = 1$. There are $2021 + 2021 - 1 = 4041$ points in total. If 4 points among these points form a non-degenerate quadrilateral, 2 points must lie on $x = 1$ while 2 points must lie on $y = 1$. But then the opposite sides are perpendicular instead of parallel, so it cannot be a parallelogram. So we are done. $\square$

Problem 40 (IMO Shortlist 2003 N7 modified). Define the sequence $\{a_n\}$ by $a_1 = 3$ and $a_{n+1} = a_n^3 + 3a_n$ for any positive integer n . Let p be an odd prime dividing a_n but not a_{n-1} , where $n > 1$. Prove that 3^{n-1} divides $p - 1$.

Solution. We work on the field $\mathbb{F}_p$. Consider the field extension $F = \mathbb{F}_p(t)$ where t satisfies $t^2 - 3t - 1 = 0$. This can be rewritten as $t - t^{-1} = 3$ (clearly $t \neq 0$), which means $t - t^{-1} = a_1$. Since

$$(u - u^{-1})^3 + 3(u - u^{-1}) = u^3 - 3u + 3u^{-1} - u^{-3} + 3u - 3u^{-1} = u^3 - u^{-3},$$

we can prove by induction that $a_n = t^{3^n} - t^{-3^n}$ for any $n \in \mathbb{N}$.

By the given conditions, we have $t^{3^n} = t^{-3^n}$ and $t^{3^{n-1}} \neq t^{-3^{n-1}}$. This means $t^{2 \cdot 3^n} = 1$ and $t^{2 \cdot 3^{n-1}} \neq 1$. Thus, the order d of t can only be 3^n or $2 \cdot 3^n$. In either case, we have $3^n \mid d$. Now, note that the degree of the field extension $[F : \mathbb{F}_p]$ can only be 1 or 2 since t is a root of a quadratic polynomial in $\mathbb{F}_p$. Thus, $F^\times$ contains $p - 1$ or $p^2 - 1$ elements. In either case, we have $d \mid p^2 - 1$. It follows that $3^n \mid p^2 - 1 = (p - 1)(p + 1)$.

We now show that $p \equiv 1 \pmod{3}$. Indeed, as $a_n = a_{n-1}(a_{n-1}^2 + 3)$, we have $p \mid a_{n-1}^2 + 3$. This shows -3 is a quadratic residue modulo p (note that $p \neq 3$ as $3^n \mid p^2 - 1$). By the Legendre symbols and the quadratic reciprocity law, we have

$$1 = \left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{3}{p}\right) = \left(\frac{p}{3}\right),$$

which proves $p \equiv 1 \pmod{3}$. Therefore, $3 \nmid p + 1$, and we finally obtain $3^n \mid p - 1$. $\square$